ZZ-M-46c
July 8, 1966

SUPERSEDING
Fed Spec ZZ-M-46b
June 20, 1962

FEDERAL SPECIFICATION

MAT, FLOOR (LINK-TYPE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies

1. SCOPE AND CLASSIFICATION

1.1 **Scope.** This specification covers floor mats of the link-type for use in corridors and entryways

1.2 Classification

1.2 1 *Classes and styles* The mats shall be of the following classes and styles, as specified (see 6.2)

Class·
- 1—Rubber, long link metal plate reinforcement
- 2—Rubberized fabric, long link
- 4—Rubber, short link
- 5—Plastic, short link

Style
- A—Mats without nosing
- B—Mats with end nosing
- C—Mats with both end and side nosing

1 2.2 *Sizes* Mats covered by this specification shall be furnished in sizes listed in table I, as specified (see 3 4).

2. APPLICABLE DOCUMENTS

2.1 **Specifications and standards.** The following specifications and standards of the issues in effect on date of invitation for bids, or request for proposal form a part of this specification to the extent specified herein:

Federal Specifications:

T-T-911—Twine, Jute.
QQ-S-781—Steel, Strapping, Flat.
CCC-C-467—Cloth, Burlap, Jute (or Kenaf).

Federal Standards:

Fed. Std No 123—Marking for Domestic Shipment (Civilian Agencies)
Fed Test Method Std. No 501—Floor Coverings, Resilient, Nontextile Sampling and Testing.
Fed Test Method Std No 601—Rubber Sampling and Testing

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U S Government Printing Office Washington, D C 20402

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston New York Washington, D C Atlanta Chicago Kansas City Mo Dallas Denver, San Francisco, Los Angeles and Seattle Wash

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies)

Military Standards

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109—Quality Assurance Terms and Definitions.
MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issues in effect on date of invitation for bids shall apply.

Official Classification Committee (OCC)

Uniform Freight Classification Rules

(Application for copies should be addressed to the Official Classification Committee 1 Park Ave at 33rd St., New York, N. Y., 10016.)

American Trucking Associations, (ATA) Inc.:

National Motor Freight Classification Rules.

(Application for copies should be addressed to the American Trucking Associations, 1616 P St N W, Washington, D C, 20036.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Links.*

3.1.1.1 *Class 1* The links of class 1 mats shall be made from a compound of natural rubber, synthetic rubber, or a suitable mixture thereof; shall be free of objectionable odor; and shall contain no material that may stain, injure, or mar the appearance of the floor.

3.1.1.2 *Class 2* The links of class 2 mats shall be made of super-imposed layers of rubberized fabric. The rubberized fabric shall contain not less than three plies of fabric. The links shall be free of objectionable odor and shall contain no material that may stain, injure or mar the appearance of the floor.

3.1.1.3 *Class 4* The links of class 4 mats shall be made from a suitably plasticized virgin polyvinyl chloride, or a compound of natural rubber, synthetic rubber, or a suitable mixture thereof. The links shall be free of objectionable odor, and shall contain no material that may stain, injure, or mar the appearance of the floor.

3.1.1.4 *Class 5.* The links of class 5 mats shall be made from a suitable plasticized virgin polyvinyl chloride that is free of objectionable odor, and contains no material that may stain, injure, or mar the appearance of the floor.

3.1.2 *Nosings* The nosings for mats, styles B and C, shall be made of the materials specified for links, and shall be tapered to prevent tripping The construction shall ensure that no metal is exposed on the wearing surface of the nosing

3.1 2.1 End nosings shall be reinforced with a metal strip entirely encased in rubber or plastic to prevent the corners from curling.

3.1.2.2 Side nosings shall be sufficiently flexible to permit rolling up of the mat without cracking, tearing, or excessive distortion of the nosing

3.1.3 *Cross wires* The cross wire shall be of corrosion-resistant or galvanized spring-steel wire The wire shall be 0 0915 inch to 0 13 inch in diameter.

3 1.4 *Metal plates* The plates shall be steel plated with cadmium or zinc, unless brass is specified in the invitation for bids (see 6.2) The plates may be shaped like the links with which they are used Hole perforations shall be suitable for the size of wire used

3.1.5 *Closed sleeve or ferrule classes 1 and 2* The closed sleeve or ferrule for joining the wire shall be carbon steel, plated with cadmium or zinc, unless brass is specified in the invitation for bids (see 6.2)

3.1 6 *Metal exposure classes 4 and 5* No metal shall be exposed on the top or bottom surfaces of the mat

3.2 Links

3.2.1 *Class 1.* Links may be of any length up to 4 inches as long as the mat remains flexible. The width shall be 1/2 ± 1/32 inch and the thickness shall be 1/2 inch ± 1/16 inch at the thickest point. The ends shall be semicircular and concentric with the cross-wire holes. The links shall have line contact with the floor at not less than three approx-

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imately equidistant points. The wearing surface of the links shall be corrugated at right angle to the length of the links.

3.2.2 *Class 2.* Links may be of any length up to 4 inches as long as the mat remains flexible and easy to roll. The width shall be 1/2 ± 1/16 inch. The ends shall either be square or semicircular and shall be concentric with the cross-wire holes. The holes shall be approximately 1/8 inch in diameter. The layers of fabric shall be substantially perpendicular to the wearing surface. The wearing surface may be corrugated.

3.2.3 *Classes 4 and 5.* The link shall be the removable link type, reversible. The links shall be extruded or injection molded uniformly on both sides. Each link shall have uniform holes for supporting the links longitudinally on the wire framework. The links shall have line contact with the floor at three or more approximately equidistant points. The wearing surface of the links shall have at least three corrugations on each side and each corrugation shall be tapered. the corrugations shall be at right angles to the length of the links. The number of links per square foot shall be not less than 120.

3.3 Color.

3.3.1 *Class 1.* Class 1 shall be brick red and black. The red links shall be used to form a red diamond in the center of the mat.

3.3.1.1 The number of red links used to form the red diamond in class 1 mats, as designated in table I, shall be not less than 8 percent and not more than 12 percent of the total number of links in the mat.

TABLE I. *Mat sizes*

Mats	Stock sizes[1]	Special sizes
Class 1:		
	14 by 24 inches	Widths up to 48 inches by lengths up to 96 inches.
	17 by 28 inches	
Class 2:		
	18 by 29 inches	Widths up to 48 inches by lengths up to 96 inches.
	18 by 36 inches	
	24 by 36 inches	
Class 4:		
	14 by 24 inches or 15-1/2 by 22-1/2 inches	Widths up to 8 feet.
	18 by 30 inches	
	18 by 36 inches	
	24 by 36 inches	
Class 5:		
	17 by 28 inches or 17-1/2 by 27 inches	Any length practical for handling overall dimensions.
	18 by 30 inches	Note. Reinforced nosing on ends only; tubing or nosing on sides.
	18 by 36 inches	
	21-1/2 by 32 inches	
	24 by 36 inches	

[1] The above sizes are approximate and the tolerance permitted is governed by the tolerances permitted for the size links in 3.2.1 and 3.2.2. The nosing is included in the measurement of the overall dimension.

3.3.2 *Class 2.* The color of class 2 mats shall be such as are naturally produced by the edge surfaces of the rubberized fabric either dark or light. Unless otherwise specified (see 6.2), a mixture of dark and light links may be used in the mat provided that a mat of suitable appearance is produced.

3.3.3 *Classes 4 and 5.* Classes 4 and 5 mats may be made up in black, brick red, grey brown light green. dark green, orange white, and beige, as single colors or in combination of colors, in any pattern or design.

3.3.3.1 Mat nosings of tubings of classes 4 and 5 mats shall be black, brick red, light brown, or grey, as specified in the invitation for bids.

3.4 Sizes and shapes. Classes 1, 2, 4, and 5 mats shall be furnished in stock sizes listed in table I, as specified: classes 1, 2, and 4 mats may be furnished in special sizes designated in table I as specified in the invitation for bids (see 6.2). When special sizes or shapes are specified, the dimensions specified shall be followed as closely as practicable but shall not be exceeded.

3.5 Construction.

3.5.1 *Class 1.* The links shall be arranged in single thickness in alternate parallel

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rows, without spacers between the links except on the ends of the mats. The links shall be secured together with cross wires and shall be placed on the wires in a manner to prevent lateral movement of the links due to traffic The mat shall be so constructed as to remain flat when laid and be readily rolled The cross wires shall fit the perforations in the links without excessive play The ends of the cross wires in the way of outside links shall be bent at right angles parallel with the links. When the mats are 37 inches or more in width, a metal plate shall be inserted over the wire on each side of the mat, i.e., on the side where the wires are bent at the beginning of the mat and on the other side where the mat has been closed as an additional strengthening The mats shall be reversible

3.5.2 *Class 2* The design of class 2 mats shall be the same as class 1 mats except that the links shall be arranged in either single or double thicknesses and the spacers shall be compounded from natural rubber synthetic rubber or a mixture of the two

3.5.3 *Classes 4 and 5* Each link shall have uniform holes for supporting the corrosion-resistant wire framework The wire framework shall extend the total width of the mat except nosing and shall be spaced at least every 1-1/16 to 1-1/8 inch between total length of mat except nosing Supporting wires shall be hooked into the side nosing at each end of the supporting wires in such a manner that they can be easily removed and reused or shall be attached with fasteners and metal screws On all matting there shall be solid or reinforced rubber nosing or round tubing There shall be reinforced nosing on the ends of the mat and tubing or nosing on the sides of the mat. For class 4, the reinforced nosing shall consist of a steel rod entirely surrounded by vulcanized rubber. For class 5, the reinforced nosing shall consist of a steel rod entirely surrounded by polyvinyl chloride.

3.6 **Weight.** The weight of the mats shall not exceed 2-3/4 pounds per square foot

3.7 **Accelerated aging.** When tested in accordance with 4.3.3, the links shall not become brittle nor tacky and shall not show any indication of deterioration

3.8 **Volatility, class 5.** When tested in accordance with 4.3.3 the loss of volatile material shall not exceed 0.5 percent

3.9 **Workmanship** The floor mats shall be free from embedded foreign material, cracks, tears, or objectionable odor and shall conform to the quality assurance provisions as specified herein

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements

4.1.1 *Terms and definitions* Terms and definitions for the examinations and tests described herein shall be in accordance with the definitions of MIL-STD-109

4.2 **Sampling for inspection and acceptance** Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105 except where otherwise indicated For purposes of sampling an inspection lot shall consist of all end items of the same class and style made with the same components and submitted for acceptance and delivery at one time.

4.2.1 *Component and material inspection.* In accordance with 4.1, the supplier is responsible for insuring that components and materials used are manufactured, tested, and inspected in accordance with the re-

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quirements of referenced subsidiary specifications and standards to the extent specified, or, if none, in accordance with this specification In the event of conflict, this specification shall govern.

4.2.2 *Inspection levels and AQL's* The acceptable quality levels, (AQL's), expressed in defects per 100 units, and inspection levels, shall be as follows·

Examination paragraph	Inspection level	AQL
4.3.1	II	4.0
4.3 2	S-2	4 0
4 3 3	S-2	4 0

4.3 End item inspection

4.3 1 *Visual examination* Examination of the end item shall be in accordance with the classification of defects and AQL's set forth herein The lot size shall be expressed in units of mats as applicable for the purpose of determining the sample size in accordance with the provisions of MIL-STD-105 The sample unit for this examination shall be one completely fabricated mat

TABLE II *Material composition*

Examine	Defect
Material, construction and workmanship	Not color specified Contains blisters, (greater than 1 16 inch diameter or more than a cluster of 4 blisters contained in an area of 1 square inch), cracks imbedded foreign material or fibers Has an objectionable odor Material not fabricated as specified Not corrugated as specified

4.3.2 *Dimensional examination* Examination shall be made for defects in dimensions The lot size shall be expressed in units of mats The sample unit shall be one completely fabricated mat. Any deviation from specified widths and thicknesses shall be considered a defect. Any length less than specified or indicated shall be considered a defect

4.3 3 *Testing of the end item.* Tests shall be performed for the characteristics specified in table III The lot size shall be expressed in units of mats. The sample unit shall be one mat. The inspection level to determine the sample size shall be S-2 of MIL-STD-105 and the AQL shall be based on 4 0 defects per 100 units

TABLE III *Test requirements*

Characteristic	Req ref	Test method
Material	3.1 1	Certificate of compliance
Link width and thickness	3 2	(a) Method 2111
Weight ..	3 6	(b) Method 7611
Accelerated aging	3 7	(a) Method 7221 or 7241
Volatility	3 8	(b) Method 9211

(a) Fed Test Method Std. No 601
(b) Fed Test Method Std. No 501

4.4 **Inspection of preparation for delivery.** The packing and marking of the mats shall be inspected to determine conformance with the applicable requirements of section 5 The packing and marking requirements of section 5 identified to referenced specifications and standards shall be inspected in accordance with the applicable requirements of these documents The packing requirements of section 5, not identified to referenced specifications. shall be examined for the defects listed in table IV

TABLE IV *Classification of packing defects in accordance with MIL-STD-105, based on an AQL of 4 0 percent defective*

Categories	Defects
Major	
101	Mats not packed in quantity specified
102	Weight of bundle exceeds limit specified
103	Ends of bundles not closed as specified
104	Bundle not reinforced with strapping as specified
105	Strapping not type or class as specified.
106	Size of strapping less than specified.
107	Size or class of mat per bundle not as specified

5. PREPARATION FOR DELIVERY

5.1 **Packing.** Packing shall be level A, B, or C as specified (see 6.2).

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5.1.1 *Level A.* Stock size mats shall be packed in bundles of six. Special size mats shall be packed in bundles not to exceed approximately 200 pounds per bundle, except when the weight of the individual mat would exceed this limitation. Bundles shall be wrapped in cloth conforming to CCC-C-467, the open ends of which shall be securely closed by sewing with twine of 2- or 3-ply jute conforming to the requirements of T-T-911, type B. In lieu of sewing, the ends may be closed with 6-inch wire ties of 15-gage soft iron or steel with one-half inch diameter formed eye at each end. Each bundle shall then be reinforced with flat steel straps conforming to the requirements of QQ-S-78, type I, class B; the dimensions of the strap shall be not less than 3/4-inch by 0.015-inch. One steel strap shall be positioned girthwise approximately 4 inches from each end of the bundle. A third strap shall be placed equidistant between the two end straps. Not more than one size or class of mat shall be packed in a single bundle.

5.1.2 *Level B.* Mats shall be packed in bundles of six. These bundles shall be wrapped in burlap cloth and the open ends shall be securely closed by sewing with twine. In lieu of sewing, the ends may be closed with 6-inch wire ties of soft iron or steel with a one-half-inch diameter formed eye at each end. Each bundle shall then be reinforced with flat steel straps conforming to the requirements of QQ-S-781, type I; the dimensions of the strap shall be not less than 3/4-inch by 0.015-inch. One steel strap shall be positioned girthwise approximately 4 inches from each end of the bundle and a third strap shall be placed equidistant between the two end straps. Not more than one size of a class of mat shall be packed in a single bundle.

5.1.3 *Level C.* The mats shall be packed in a manner that will insure arrival at destination in satisfactory condition, and be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification Rules or National Motor Freight Classification Rules.

5.2 Marking. (See 6.2).

5.2.1 *Civil agencies.* In addition to any special marking required by the contract or order, the bundles shall be marked in accordance with Fed. Std. No 123.

5.2.2 *Military agencies.* In addition to any special marking required by the contract or order, the bundles shall be marked in accordance with MIL-STD-129

6. NOTES

6.1 Intended use. Mats, style A, are suitable for recessed floors or for locations when tripping is not a risk. Mats, style B, are suitable for runners where there is no side traffic crossing. Mats, style C, are suitable for both side and end traffic crossing. Mats, classes 4 and 5 are suitable for use when traffic includes women wearing narrow tip heels

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(*a*) Title, number, and date of this specification

(*b*) Class and style required (see 1.2.1)

(*c*) Whether stock or special size is required (see 1.2.2 and 3.4)

(*d*) Type of metal required in link (see 3.1.4 and 3.1.5)

(*e*) Design and color, if required (see 3.3)

(*f*) When a mixture of dark and light links may not be used (see 3.3.2).

(*g*) Level of packing required (see 5.1).

(*h*) Marking required (see 5.2).

6.3 Class 3 of ZZ-M-46b is deleted because there is no requirement for it.

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MILITARY CUSTODIANS:

Navy—YD
Air Force—69

Review Activities:

Navy—YD
Air Force—69

User Activity:

Navy—MC, CG

Preparing Activity:

Navy—YD

CIVIL AGENCIES INTEREST:

GSA-FSS
NAS
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Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 5 cents each

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
ZZ-M-46c: Mat, Floor (Link-Type)

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 1A 0101-807-2000 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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